

Report on Need of Certification System for CEMS in India



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Abbreviations

ACIME	Association for the Certification of Measuring Instruments for the Environment
AETB	Air Emissions Testing Bodies
AMS	Automated Measuring Systems
AST	Annual Surveillance Test
ASTM	American Society for Testing and Materials
CAIR	Clean Air Interstate Rule
CAMD	Clean Air Markets Division
CE	Conformité Européenne
CEMS	Continuous Emission Monitoring Systems
CEN	European Committee for Normalization
CFR	Code of Federal Regulations
CSA	Canadian Services Association
EA	Environmental Agency
EA	European Cooperation for Accreditation
ECMPS	Emissions Collection and Monitoring Plan System
EPA	Environmental Protection Agency
EN	European Norms
EU	European Union
ISO	International Organization for Standardization
MCERTS	Monitoring Certification Scheme
MoEF&CC	Ministry of Environment, Forest and Climate Change
QAL	Quality Assurance Level
RATA	Relative Accuracy Testing Audits
SCS	Sir Certification Scheme
TGN	Technical Guidance Note
UBA	Umweltbundesamt

1. Introduction

Continuous Emission Monitoring System (CEMS) is an active and indispensable part of pollution monitoring in developed economies like members of European Union (EU), United States of America (USA), Japan etc. It has been extensively utilized to regulate emissions of gases and particulate matter in the atmosphere for the last 25 to 30 years. One of the most crucial factors behind the success of CEMS in these countries is the assured systems required such as for certification or quality assurance of devices, extensive testing & verification of devices by independent agencies, defined roles and responsibilities and guidelines.

India adopted the continuous emission and effluent monitoring systems two years back in 2014 when Central Pollution Control Board (CPCB) mandated major industries and common pollution control facilities to install CEMS. Irrespective to the scale of operation, continuous effluent quality monitoring has been mandated in all industries situated in Ganga basin. Now, implementation of this initiative requires all the must have systems such as certifications system, lab empanelment systems and the respective guidelines and protocols for CEMS implementation. Such systems have been available in India but not for CEMS.

Therefore, the best way to achieve this is to look at how these systems have been set-up in other countries, how existing similar systems work in India and accordingly develop a strategy to develop the required systems. This report is an attempt in this direction. It identifies the systems available in other countries like US, UK, Germany etc., existing similar systems in India and discusses how India can move ahead.

2. CEMS Device Certification

Device certification ensures that the product manufactured is reliable and works as per the laid standard. Certification is carried out by a competent authority or agency authorized to evaluate and approve the quality and functioning of the product as per the relevant national or international standard. Once the device passes all the tests specified in such standards, it gets certified. Certification gives the product recognition and quality assurance across different markets where it is recognized. This eliminates any requirements of retesting the device when it meets a new market allowing international trade runs smoothly.

As far CEMS is concerned, it is purposefully utilized to collect accurate and credible data monitored from the source. This is basis of self-monitoring and control practices like emission trading. Certification assures the quality and accuracy of the equipment which is recognized by relevant stakeholders. The countries like Germany, United Kingdom (UK), US etc. has well established CEMS framework and so their quality assurance system for CEMS. Europe has a system of CEMS certification while USA has adopted a similar system of CEMS performance check during installation. India has adopted CEMS very recently but has no indigenous system, therefore, the certified CEMS equipment which have been imported brings some credibility for its quality and

performance, but the indigenous ones cannot as they have not been tested and evaluated by any recognized agency. The success of CEMS initiative demands to develop an indigenous certification system, which can assure the credible and quality data for compliance.

2.1. Certification for CEMS in Europe

The European certification schemes include both type approval of the product (products that meet a minimum set of regulatory, technical, and safety requirements for a particular type of process), and factory audits carried out regularly to ensure that the product being made and sold is still identical to that which was tested. Certification is often regulation-driven. Table 1 shows certification schemes in various European countries.

Initially, European countries had their individual set of standards to which devices had to be accredited to. This was unfavorable to all parties involved in international trading, as it required a fresh accreditation each time products met a new market. It made international trade difficult. To solve this problem, a single accreditation that would be considered acceptable to all countries and effective in terms of quality across Europe was required. Hence, the standard BS EN – 15267 was structured to certify CEMS devices and to give them international recognition.

This standard is part of the Monitoring and Certification Scheme (MCERTS) in the United Kingdom. MCERTS is a part of the Sira Certification Service (SCS) under by Canadian Standards Association (CSA) Group Testing UK Ltd. is the certification body which runs MCERTS scheme in the UK.

UKAS has accredited SCS as per ISO/IEC 17000 series of conformity assessment standards to undertake the product and personnel certification. All device testing under MCERTs is carried out by laboratories or agencies accredited to ISO 17025 (accreditation for testing and calibration). CSA examines the results of the laboratory tests and field tests using a group of independent experts (Certification Committee). The requirements for certification of CEMS under MCERTS are covered in BS EN 15267-1, the requirements for the manufacturer's quality management system for manufacturing and design control are covered in BS EN 15267-2. BS EN 15267– 3 covers the performance testing processes to be carried out in laboratories, field tests.

In Germany, TUV is authorized by UBA to carry the testing and certification for of CEMS device. Germany's process for CEMS device certification falls on the same lines as that of the standard EN 15267.

Table 1: European systems for CEMs certification/type approval¹

Country	Certification scheme	Mandatory	Date of beginning	Salient features
UK	The Environment Agency of England and Wales (EA) Monitoring Certification Scheme MCERTS	No	1998	• De facto, all CEM manufacturers apply • Verifies the compliance with the performance standards and audits the production plant • Data generated by 5 labs is accepted: 2 UK labs, 1 French, 1 German and 1 Italian
France	Association for the Certification of Measuring Instruments for the Environment ACIME	No	2003	• 2 labs in France are entitled to do the tests • Data from other labs is accepted (e.g. TUV) Only 3 vendors have been certified; because of the poor success, the certification may be stopped.
Germany	UBA (Umweltbundesamt) Type Approval scheme	Yes (to sell in Germany)	1975	• Verifies the performance of the device, i.e. its suitability to meet minimum requirements defined to be in line with the German regulations. • Only data generated by 5 German labs are accepted
European Union	CEN (European Committee for Standardization) standard: Certification scheme	No		The standard is still under development

Source: L.Sophie et al, JRC Scientific and Technical Reports, 2007, Environmental Technologies Verification Systems, p57

2.1.1. Certification process

CEMS devices are certified only if certification is performed according to standard BS EN 15267– 1, the manufacturer adheres to the requirements of BS EN 15267-2 and the measuring system has been tested as per the standards mentioned in BS EN 15267- 3.

EN 15267 Part 1 - General principle: This specifies the general principles for the product certification of CEMS for monitoring emissions from stationary sources and ambient air quality. The product certification consists of the sequential steps- Performance testing of a CEMS, Initial assessment of the CEMS manufacturer's quality, Certification and Surveillance. The flow chart below (figure 1) represents the process of MCERTS certification.

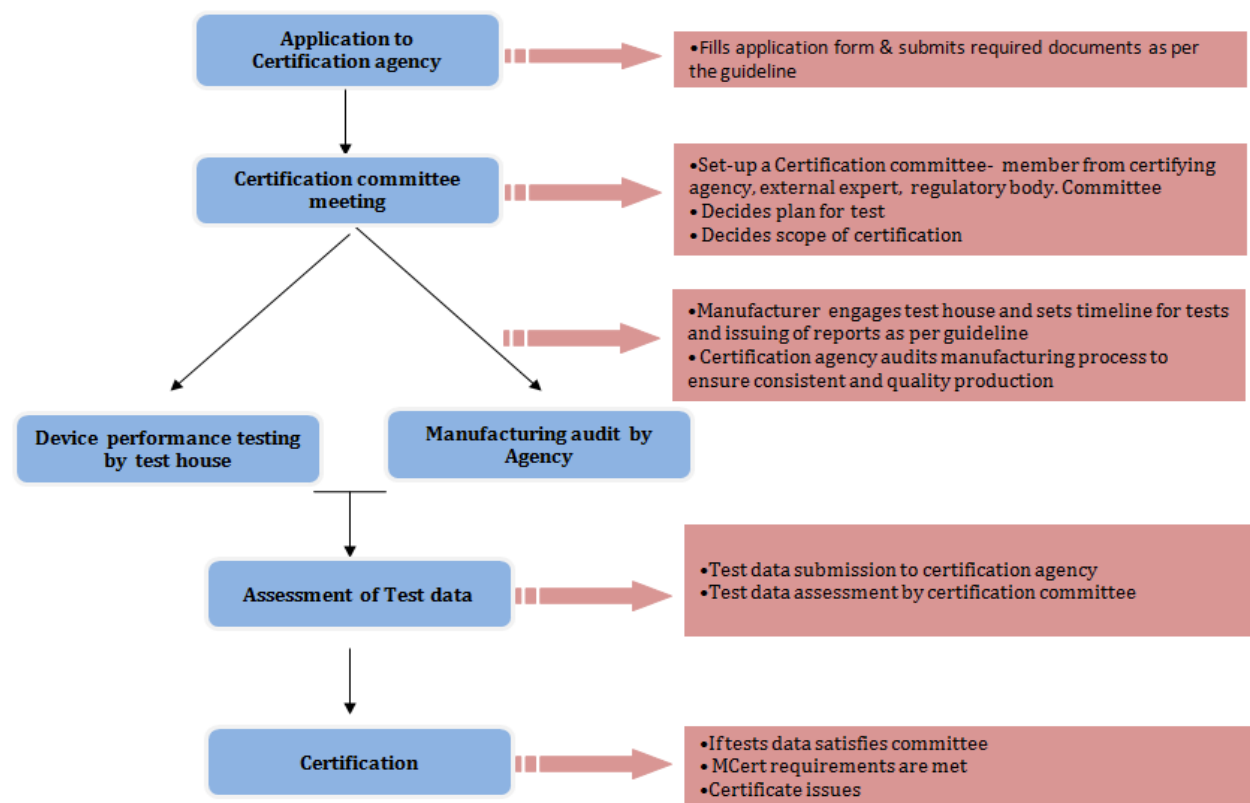


Figure 1: Certification for CEMS in Europe

Part 1 of BS EN 15267 lays down the responsibilities of all the bodies that are expected to be a part of the accreditation process namely, the manufacturer, the testing laboratory where devices shall be tested and competent authorities and certification bodies, collectively given the name 'relevant bodies'. Following certification, the relevant bodies ensure that surveillance of the continued manufacturing and performance of certified CEMS is carried out periodically.

In some CEN member states the competent authority delegates the responsibility for approval of CEMS to a Certification Body accredited to EN 45011 (replaced by EN ISO/IEC 17065) by national accreditation agencies/bodies. These approaches have built up over many years and reflect the different administrative and legal arrangements that exist in different member states. In order to recognize these different approaches, EN 15267-1 uses the collective term “relevant bodies” when referring to competent authorities and certification bodies.

In Germany, guidelines published by Federal Ministry of Environment or Association of German Engineers (VDI -Verein Deutscher Ingenieure) or DIN EN Guidelines are followed for carrying out testing to obtain certification. These guidelines discuss suitability testing of measuring and evaluation systems for continuous emission measurement; installation calibration and maintenance of continuously operating measuring and evaluation devices; and evaluation of continuous emission monitoring. Apart from their nomenclature, these guidelines are built on the same lines as that of

the standard EN 15267.

VDI has worked out testing plans for carrying out suitability tests and published them as VDI Guidelines. Suitability is tested at the expense of the manufacturer and by testing institutes/ laboratories accredited for suitability testing as per DIN EN ISO/IEC 17025. These institutes are required to fulfill licensing requirements and have five years of experience in the same area of suitability testing. Testing institutes/laboratories accredited through the accreditation system of ILAC are also considered as equals² Since there are different stakeholders involved in this, clear roles and responsibilities have been laid for each of them (Table 2).

Table 2: Roles and Responsibilities of Manufacturer, Test Laboratory and Relevant Body

MANUFACTURER	TEST LABORATORY	RELEVANT BODY
Submit at least two complete and identical CEMS and information for testing	Evaluates CEMS conformity with performance criteria	Provides guidance on arrangements and requirements for certification to manufacturer and test laboratory
Establishes, maintains and operates a quality management system as per EN 15267-2	Accredited to EN ISO/IEC 17025	Accredited to EN 45011 if the relevant body is a certification body
Ensures quality assurance and control of manufacturing such that certified CEMS continue to meet applicable performance criteria	Advises manufacturers on suitability of CEMS for different measuring ranges	Have appropriate procedures for certification, if it is an unaccredited competent authority
Controls and assesses design changes and keeps detailed records on that	Applies for the certification of the AMS at the relevant body	Assesses test reports and determine, whether test laboratory is appropriately accredited to carry out tests
Records methods and results of re-testing, if the CEMS requires partial or complete re-testing	Evaluates any design changes to the CEMS if requested by the manufacturer	Verifies that manufacturer has an EN 15267-2 conforming quality management system
Notifies the test laboratory and the relevant body of changes to the CEMS	if original certification is affected, relevant body is notified if retesting is required, advise manufacturer and relevant body	Cooperates as required with the relevant national competent authority
		Issues certificates with an appropriate scope in (English/French/German)
		Adds the CEMS to the records
		Ensures post certification surveillance is periodic.

BS EN 15267 Part 2 - Quality Management (QM) and audit scope

This standard specifies the requirements for QM system of the manufacturer and post certification surveillance for manufacturing process. Since, CEMS may undergo design changes during product life, it is important to ensure the changes don't alter its performance beyond standards. These design changes may be of Type 0 (Changes having no measurable influence to the performance of the CEMS), Type 1 (Changes with a possible influence, but for which can be proved by test, that the influence is not significant) and Type 2 (Changes with significant influence to the CEMS). Additional tests are always required, if type 2 changes are implemented.³ The manufacturer must keep records and evaluations on any modification (software and hardware) of the certified CEMS.

In order to control the changes, EN 15267-2 specifies the requirement of manufacturer's quality management system, initial assessment of manufacturer's production control and continuous surveillance of the effect of CEMS' design changes on its performance.

BS EN 15267 Part 3 - Performance Standards and Test Procedures

This European Standard defines performance standards and test procedures for CEMS that measure gases, particulate matter and flow of waste gases from stationary sources. It provides detailed procedures that cover QAL1 requirements of the quality assurance standard EN 14181. These procedures are a combination of laboratory tests and field tests. Laboratory tests last between 4 – 8 weeks whereas field tests are carried out for a minimum duration of 3 months.

2.2. Certification for CEMS in the USA

United States Environmental Protection Agency (USEPA), under Clean Air Markets Division⁴ (CAMD), operates several market based programs to improve the quality of air. These include the Acid Rain Program and Clean Air Interstate Rule (CAIR). Industries and other (emission) sources that are regulated by these programs follow the monitoring regulations in 40 CFR Part 75 which requires continuous monitoring and reporting of sulfur dioxide (SO₂), carbon dioxide (CO₂) and Nitrogen oxides (NO_x) emissions. These pollutants are monitored using CEMS.

40 CFR Part 75 Sub part C describes the process for certification and re-certification of CEMS, its quality assurance and quality control (QA/QC) requirements, defines 'out-of-control' periods, and requires bias adjustment of data from SO₂, NO_x, and flow monitors. Its "Appendix A" describes CEMS installation and certification test procedures, and provides performance specifications for the CEMS and explains how to set the span and range of CEMS and Appendix B describes the required on-going CEMS quality assurance tests and procedures for CEMS, and includes rules for data validation⁵.

There are a number of monitoring methods available based on the composition of emissions produced from fuel consumed (for example, combustion of very low level

sulfur fuels doesn't require CEMS). CEMS are used to provide the emissions data unless the unit (source) qualifies to use one of the alternate monitoring methods specified in the rule. The selected monitoring methodology for each unit must be approved by USEPA through a certification process. Once the methodology has been approved and the required monitoring systems are certified, the recording and reporting of emissions data begins.

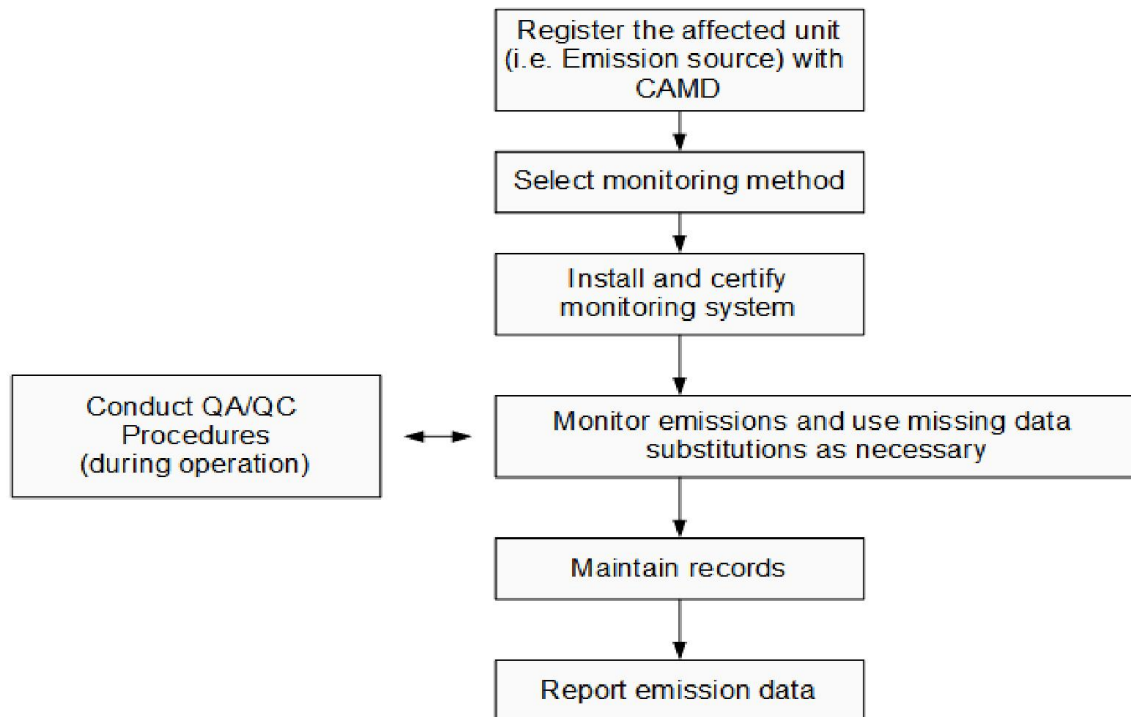


Figure 2: Overview of Part 75 Monitoring Requirements⁶

2.2.1. Certification process

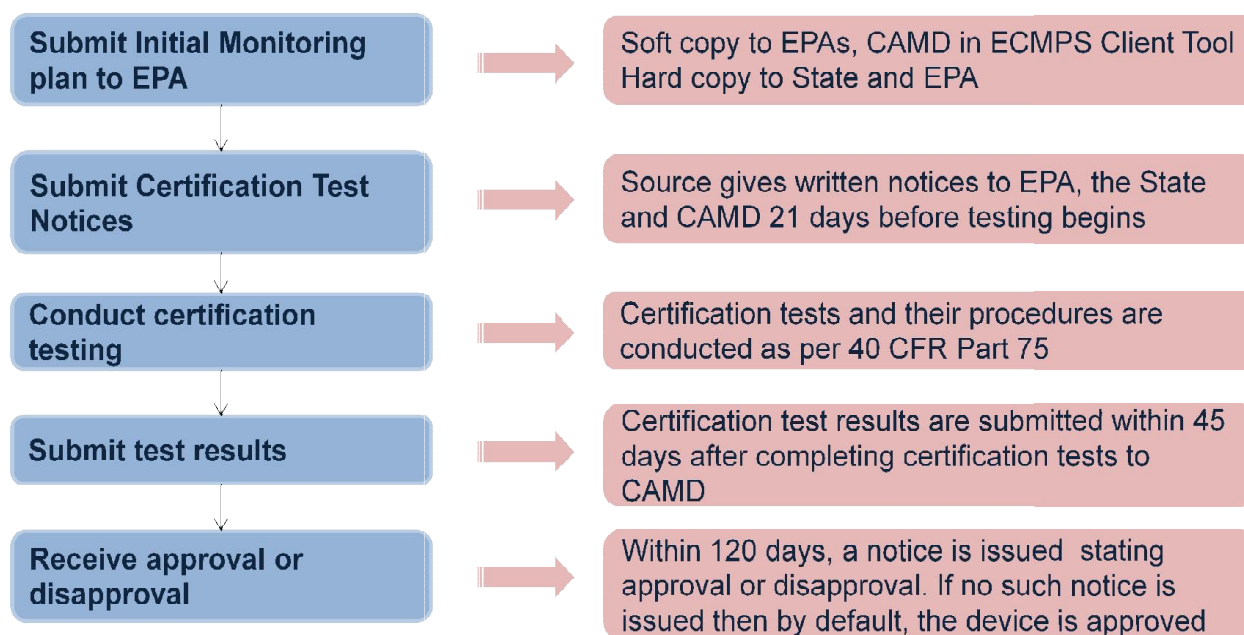
The USA doesn't have a CEMS device certification process similar to Europe. Rather the device is checked for performance assurance during the installation at site. The responsibility of getting the CEMS certified lies with the owner/operator.

The applicant submits an initial monitoring plan to USEPA and the State at least 21 days before certification testing begins. The electronic copy includes information like device specifications, monitoring method whereas; the hard copy includes information which may be incompatible with the ECMPS (Emissions Collection and Monitoring Plan System) Client Tool. The source provides written notice to CAMD, EPA and the State at least 21 days before testing begins. The certification testing is carried as per the procedures specified in 40 CFR 75 Section 6 of Appendix A. These tests include 7-days calibration error tests, linearity check, RATA, bias test, cycle time test, flow meter accuracy test, four load NO_x emission testing and heat input measurement, NO_x emission testing at one or more unit test and DAHS verification etc.⁷

Once certification tests are completed, the application for certification is electronically to CAMD and the documents that are incompatible to electronic tools goes as hard copy to state and USEPA Region within 45 days. The hard copy of application consists of certification test report and changes (if any) in the monitoring plan whereas the electronic copy consists of a complete, updated monitoring plan and the results of the certification tests.

If the application is incomplete in any aspect, sufficient time is given to procure the missed out details. An appropriate reviewing agency (USEPA for the Acid Rain Program and State/local agency for the CAIR programs) issues a notice of approval or disapproval of certification in 120 days. If no such notice is issued, the device is deemed certified by default.

Among the test procedures, all Relative Accuracy Testing Audits (RATAs) for CEMS under Part 75 are carried out by an Air Emissions Testing Body (AETB) that has been certified by demonstrating conformance to ASTM (American Society for Testing and Materials) D7036-04. The operator/owner obtains AETB certification to check its validity at the time of testing.⁸



abbreviations:

CAMD – Clean Air Markets Division

ECMPS - Emissions Collection and Monitoring Plan System

Figure 3: Certification for CEMS in USA

2.3. Certification for CEMS in India

CEMS is the credible and accurate system of pollution monitoring and reporting. The developed countries which have adopted CEMS have also set-up systems for device quality assurance or performance check for this. Mainly, two types of system for device quality assurance and performance check are available which have been highlighted in this report.

In India, presently such system doesn't exist that leaves enough scope for poor quality product to be used. This questions accuracy of the device and credibility of the data. Not for CEMS, but for other such equipment, Bureau of Indian Standard (BIS) is the national standards organization/agency in India which is authorized for conformity testing of product and grant Indian Standards Institution (ISI) certification for the quality, safety and reliability of the product. Some other private organizations such as British Standard Institutions (BSI) groups, United Registrar of system (URS) certification etc. are involved in product certifications like CE (Conformité Européene), BSI Kitemark etc. for certain products. As of now, none of these agencies carry certification for CEMS in India.

Table 3: Gaps to bridge

S. No.	Gaps	India	Germany	UK	USA
1.	Specific guidelines for CEMS testing and calibration laboratories	Nil	Yes	Yes	Yes
2.	Suitability of CEMS	Nil	VDI 4203-1	QAL 1 EN14181	EPA
3.	Certifying body for CEMS device	Nil	German Federal Environment Agency (UBA)	CSA	

2.3.1. Product certification by BIS

BIS operates a product certification scheme under the Bureau of Indian Standards Act, 1986. Under this scheme, BIS grants licenses to manufacturers from all industries.⁹

The BIS product certification scheme is essentially voluntary, and is largely based on ISO/IEC Guide 28 which provides general rules for third party product certification system for determining the conformity of a product with specified requirements through initial testing of samples of the product, assessment and surveillance of the involved quality system, and surveillance by testing of product samples taken from the factory or the open market, or both.¹⁰

All BIS certifications are carried out in accordance with Indian Standards, which are amenable to certification. A large number of operational elements of the BIS product certification scheme correspond with the requirements of ISO/IEC 17065. The BIS employs competent engineers and scientists to cater to all its fields of operations. They

are trained into evaluation and assessment techniques to a high degree of professional competence. All preliminary and surveillance inspections are carried out by qualified personnel only. BIS has set up eight laboratories in different cities of India for testing samples of products taken during preliminary and surveillance operations. In addition, independent laboratories having demonstrated ability and a quality system complying with IS/ISO/IEC 17025 have been recognized for testing of samples.

The process of obtaining a license from BIS to demonstrate quality of the product is summarized ahead (figure 4).

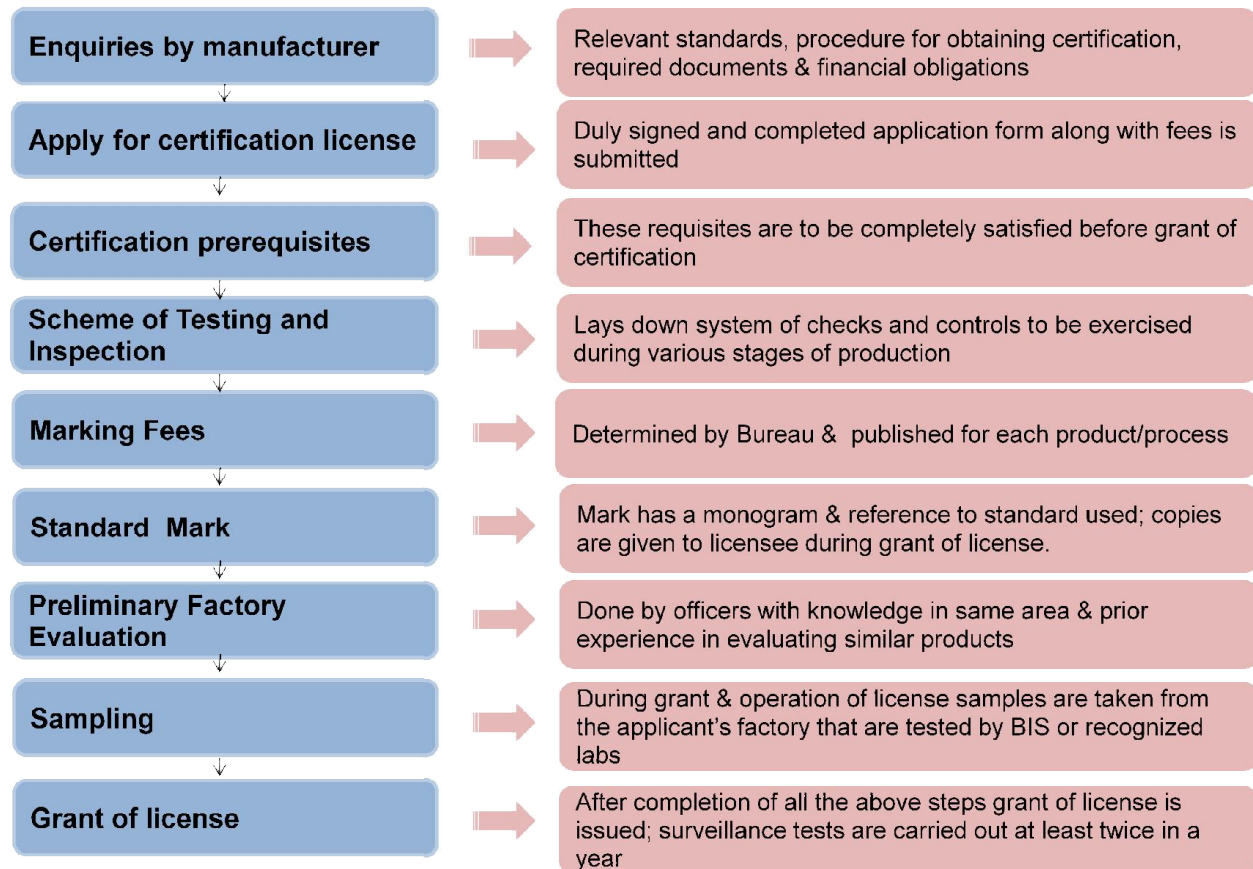


Figure 4: Product certification by BIS¹¹

Another well recognized product certification that is carried in India is the CE marking. This marking is a great promoter of international trade. Goods and products bearing this marking is acknowledged under EU and EEA

2.3.2. CE product certification

CE marking under BSI gives products legal access to the European Market and declares that the product complies with all applicable European Regulations and Directives. The responsibility of getting the product certified lies with the manufacturer. The following diagram (figure 5) summarizes the process of getting this recognition.¹²

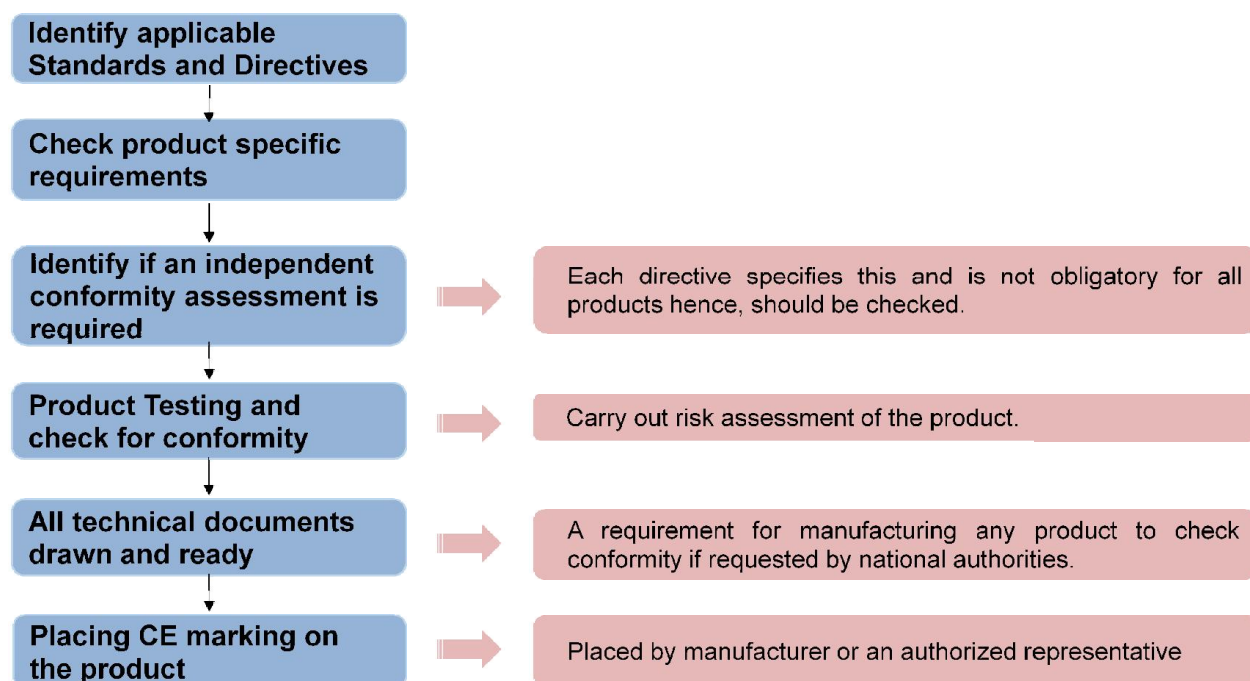


Figure 5: CE product certification process

3. Recommendation for a CEMS certification system in India

Like other certification agencies in Europe and US, the above discussed systems have well drafted guidelines and working methods which ensure recognitions of certified product. A similar quality control and quality assurance of CEMS is needed. An indigenous system of analyzer and complete device (analyzer and other accessories) certification needs to be developed.

- The government, precisely MoEF&CC or CPCB, needs to take initiative for this and approach the competent agencies and develop strategy for developing such a system in India. Keeping the timeline in mind, mandating the certification for domestically produced CEMS within a given timeframe will be required.
- BIS appears to be one of the most suitable Indian organizations for developing certification system for CEMS. It functions in line with international standards ISO/IEC 17021 that assures the competence, consistency and impartiality of bodies providing audit and certification of all types of management systems. However, it needs to get accredited and adopt the international standard for certification of CEMS.
- TUV, BIS etc. are other competent agencies available in India which have been successfully carrying this task in Europe and other countries for many decades.

They have expertise and set-ups in India which can be upgraded if required. MoEF&CC or CPCB can approach them and develop the strategy.

- These agencies may be asked to prepare a proposal and plan for setting up such a facility in a given time line.
- Since these agencies will have to invest in developing adequate infrastructure, Government's support and assurance will be encouraging to them.
- These agencies will have to be authorized by the Indian government or competent bodies for the certification job.
- A set of performance standards needs to be developed for certification against which the CEMS will be tested. For instance, standards for different tests can be developed in the same line that exists in Europe. These performance standards are well established for decades in Europe and have been followed in many countries.
- The certification agencies will be required to obtain international accreditation and follow established standards such as EN14181 (in Europe) and ISO/IEC 170256 if the certification has to ensure international traceability.
- Skill development of all the stakeholders viz. competent agencies, laboratories, regulators and industries are crucial which shall be accomplished with sufficient trainings.

Setting-up a certification system may require considerable time and effort, therefore, meanwhile following temporary quality assurance and quality control measures can be taken to ensure the quality of the CEMS device:

- The industries which are yet to install or in process of replacing the CEMS, should be advised to use certified devices only.
- In case of indigenous devices, they should be tested by competent and authorized laboratories against their performance standards during installation, in the same line as happens in USA. CPCB, being the competent agency should decide on required tests to be carried out for performance testing which can assure the quality of the device.

CPCB has a leading role in implementation of CEMS in India, therefore it has opportunity and responsibility to take lead towards developing a certification system in place. It may recommend and work with competent authority to take action in this direction. Many countries have adopted CEMS during last 3-4 decades and have

developed such systems. India has opportunities to learn from their experience and work with competent agencies and experts to set-up the system. A time bound action plan is required.

4. References

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